Geometric Sequence Formulas — Answer Key

Part 1 Recursive Formulas

1. $r = 3;\ a_1 = 2,\ a_n = 3 \cdot a_{n-1}$
2. $r = 2;\ a_1 = 1,\ a_n = 2 \cdot a_{n-1}$
3. $r = 2;\ a_1 = 5,\ a_n = 2 \cdot a_{n-1}$
4. $r = 4;\ a_1 = 3,\ a_n = 4 \cdot a_{n-1}$
5. $r = 3;\ a_1 = 4,\ a_n = 3 \cdot a_{n-1}$
6. $r = 5;\ a_1 = 1,\ a_n = 5 \cdot a_{n-1}$
7. $r = 2;\ a_1 = 6,\ a_n = 2 \cdot a_{n-1}$
8. $r = 10;\ a_1 = 2,\ a_n = 10 \cdot a_{n-1}$
9. $r = 2;\ a_1 = 7,\ a_n = 2 \cdot a_{n-1}$
10. $r = 3;\ a_1 = 3,\ a_n = 3 \cdot a_{n-1}$
11. $r = -2;\ a_1 = 1,\ a_n = (-2) \cdot a_{n-1}$
12. $r = -3;\ a_1 = 2,\ a_n = (-3) \cdot a_{n-1}$
13. $r = -2;\ a_1 = 5,\ a_n = (-2) \cdot a_{n-1}$
14. $r = \frac{1}{2};\ a_1 = 64,\ a_n = (\frac{1}{2}) \cdot a_{n-1}$
15. $r = \frac{1}{3};\ a_1 = 81,\ a_n = (\frac{1}{3}) \cdot a_{n-1}$
16. $r = 2;\ a_1 = -1,\ a_n = 2 \cdot a_{n-1}$
17. $r = 3;\ a_1 = -3,\ a_n = 3 \cdot a_{n-1}$
18. $r = -1;\ a_1 = 10,\ a_n = (-1) \cdot a_{n-1}$
19. $r = 5;\ a_1 = 4,\ a_n = 5 \cdot a_{n-1}$
20. $r = \frac{1}{10};\ a_1 = 1000,\ a_n = (\frac{1}{10}) \cdot a_{n-1}$

Part 2 Explicit Formulas

21. $a_1 = 3, r = 2;\ a_n = 3 \cdot 2^{n-1}$
22. $a_1 = 1, r = 3;\ a_n = 3^{n-1}$
23. $a_1 = 2, r = 5;\ a_n = 2 \cdot 5^{n-1}$
24. $a_1 = 4, r = 2;\ a_n = 4 \cdot 2^{n-1}$
25. $a_1 = 5, r = 3;\ a_n = 5 \cdot 3^{n-1}$
26. $a_1 = 1, r = 4;\ a_n = 4^{n-1}$
27. $a_1 = 2, r = 2;\ a_n = 2 \cdot 2^{n-1}$
28. $a_1 = 6, r = 3;\ a_n = 6 \cdot 3^{n-1}$
29. $a_1 = 1, r = 10;\ a_n = 10^{n-1}$
30. $a_1 = 3, r = 5;\ a_n = 3 \cdot 5^{n-1}$
31. $a_1 = 2, r = -2;\ a_n = 2 \cdot (-2)^{n-1}$
32. $a_1 = 1, r = -3;\ a_n = (-3)^{n-1}$
33. $a_1 = 4, r = -2;\ a_n = 4 \cdot (-2)^{n-1}$
34. $a_1 = 32, r = \frac{1}{2};\ a_n = 32 \cdot (\frac{1}{2})^{n-1}$
35. $a_1 = 27, r = \frac{1}{3};\ a_n = 27 \cdot (\frac{1}{3})^{n-1}$
36. $a_1 = -2, r = 3;\ a_n = -2 \cdot 3^{n-1}$
37. $a_1 = -1, r = 2;\ a_n = -1 \cdot 2^{n-1}$
38. $a_1 = 5, r = -1;\ a_n = 5 \cdot (-1)^{n-1}$
39. $a_1 = 7, r = 3;\ a_n = 7 \cdot 3^{n-1}$
40. $a_1 = 5000, r = \frac{1}{10};\ a_n = 5000 \cdot (\frac{1}{10})^{n-1}$

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